

Adaptive Variable Resolution 2.5D Lidar Mapping

A lidar perception system for defence ground vehicles that decides **where to spend its measurements** — and, uniquely, knows the difference between *“there is nothing there”* and *“I did not look there.”*

Same obstacle safety from roughly a third of the points. And when the budget drops, it still **cannot** miss an obstacle above a provable minimum size inside the vehicle's braking distance.

PROBLEM ID SIH26053	ORGANISATION DRDO	CATEGORY Software
THEME Smart Automation	PRIZE ₹1,00,000	GRAND FINALE December 2026

VERIFY BEFORE ANYTHING ELSE

The problem-statement listing shows **20 September 2026**; other sources say the 30th. Your college's internal hackathon date is earlier still, and it is the one that actually binds. **This plan is scheduled against the 20th.** Confirm both on sih.gov.in and with your SPOC today.

CONTENTS

- | | |
|------------------------|----------------------------|
| 01 The insight | 02 What we build |
| 03 How we prove it | 04 Checked against winners |
| 05 What we don't build | 06 The six slides |
| 07 Winner mentality | 08 The schedule |
| 09 Roles | 10 Risks |

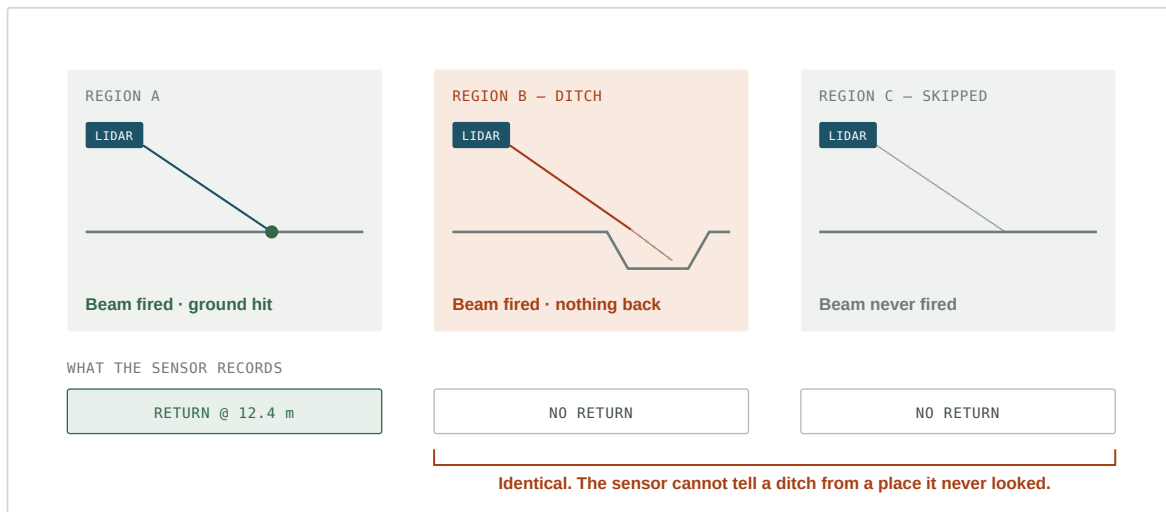
The insight the whole project rests on

This is what separates us from every other team and from every existing paper. Everything after this is execution.

Negative obstacles — ditches, craters, washouts, shell holes — are detected by the *absence* of laser returns. You see a hole because the ground stopped returning where the ground plane said it should.

Adaptive sampling deliberately creates absences. It skips regions on purpose.

Put those two facts together and a fatal ambiguity appears.



The ambiguity at the heart of the problem statement. Region B is a ditch that could destroy the vehicle; region C is ordinary ground the allocator chose to skip. Both produce exactly the same record. Naive adaptive sampling has no way to separate them.

OBSERVATION	EXPLANATION A	EXPLANATION B	EXPLANATION C
No return from this patch of ground	A ditch	Occlusion shadow	We never looked

On a road that ambiguity is a nuisance. On a UGV crossing broken terrain at speed, it is how you lose the vehicle.

WHY NOBODY HAS SOLVED THIS

All published adaptive-lidar work targets urban autonomous driving — KITTI, nuScenes, lanes, cars, pedestrians — where negative obstacles are essentially absent. Meanwhile the entire negative-obstacle literature assumes dense uniform scanning. **The two fields have never been put in the same room.**

Raysense puts them in the same room. Our allocator carries an explicit three-valued state — **OBSERVED** **UNKNOWN** **CANDIDATE_NEGATIVE** — never conflates unknown with empty, and **actively spends measurement budget to resolve dangerous ambiguity.**

That is a defensible research contribution, a safety argument a defence audience grasps instantly, and it hands us the most persuasive demo available: *watch the naive method drive into a ditch, then watch ours refuse to.*

02 – ARCHITECTURE

What we build

A **point-budget allocator**. Given a budget of B measurements per frame, where B is far smaller than a full scan, decide how to distribute them across the field of view to maximise terrain-map quality and obstacle safety.

The per-frame loop

Single pass. No sparse-then-rescan round trip — on a moving platform the world has already shifted by the time you go back.

- 1 **Predict**
Warp the map from frame $t-1$ into frame t using ego-motion. Propagate object tracks. Age every cell.
- 2 **Build the need map**
Score every angular bin: elevation gradient, staleness, predicted moving-object footprints, frontier cells, proximity to the braking corridor — and unresolved negative-obstacle candidates.
- 3 **Apply the safety floor**
A hard geometric constraint, allocated *before* anything adaptive. Never traded away.
- 4 **Allocate the remainder**
Distribute the rest in proportion to need, with a per-bin cap so it cannot collapse onto one region.
- 5 **Acquire**
Through one of three interchangeable sensor backends — this is our answer to “which lidar?”
- 6 **Update the map**
Fuse the selected points into a quadtree elevation map, subdividing where variance or gradient is high.
- 7 **Disambiguate**
Track rays, not just points. A cell is **EMPTY** only if a ray actually traversed it and returned from beyond. Everything else stays **UNKNOWN**.
- 8 **Output traversability**
Three-valued: traversable, blocked, unknown. **Unknown is never treated as safe.**

The safety floor — provable, not heuristic

To guarantee a beam strikes an obstacle of height h at range r , the angular sample spacing must satisfy:

$$\Delta\theta \leq h / r \quad \text{therefore} \quad h_{\min}(r) = r \cdot \Delta\theta$$

Inside the braking corridor we enforce $\Delta\theta \leq h_{\text{target}} / d_{\text{brake}}$ and allocate that budget first. This falls straight out of honest geometry — and it hands us a capability no other team will have: **the allocator can state the maximum safe speed for a given point budget and sensor.**

DETECTING A 20 CM OBSTACLE			
VEHICLE SPEED	BRAKING DISTANCE	REQUIRED $\Delta\theta$	FEASIBLE ON A 64-BEAM SENSOR?
5 m/s	~8 m	1.43°	Comfortably
10 m/s	~28 m	0.41°	Tight
15 m/s	~60 m	0.19°	No — cap the speed

For a defence customer, that last row is not a limitation to hide. It is the entire argument: a perception system that *knows* what it cannot see, and says so.

Three sensor backends

The first question from a DRDO electro-optics engineer will be “what sensor does this run on?” We answer it by being agnostic.

BACKEND	WHAT IT DOES	WHAT IT ACTUALLY SAVES
Steerable MEMS / OPA	Emits a real scan command	Laser energy + bandwidth + compute
Fixed-pattern Velodyne, Ouster	Masks which returns get processed	Bandwidth + memory + compute — not laser power
Replay recorded dataset	Selects a subset of a recorded full scan	This is our evaluation harness

STACK

Python with NumPy/Numba for the allocator · Open3D for point-cloud work · quadtree in NumPy, dropping to C++ via pybind11 only if latency demands · PyTorch only for a learned need map, which is a stretch goal, not core. **Keep it boring.** Judges score the result, not the framework.

How we prove it

The experiment *is* the project. Design it first, build backwards from it.

Data — two legs, deliberately

RELLIS-3D is the primary. A real all-terrain UGV in rugged off-road terrain, Ouster OS1, 13,556 point-wise annotated scans, 20 semantic classes — the closest public dataset to a DRDO ground-vehicle scenario that exists. **The full recorded scan is our ground truth;** our algorithm chooses which points it *would have acquired*. That is how we escape the obvious kill shot: “*of course it works, you wrote the simulator and the answers.*”

Simulation is the second leg, for closed-loop behaviour on a moving platform and — crucially — for **deliberately planted negative obstacles with exact ground truth**, which real datasets barely contain.

Say that two-leg rationale out loud. It turns “why simulation?” from an attack into evidence that we thought about experimental design.

Baselines — all at identical point budget

1. **Full scan (100%)** — the quality ceiling
2. **Uniform decimation** — every k -th beam. The honest naive baseline.
3. **Random subsampling**
4. **Static front-focused ROI** — the “reasonable engineer” baseline, and the one we must beat convincingly, because a judge *will* propose it
5. **Ours**

METRIC	WHY IT EARNS ITS PLACE
Elevation MAE / RMSE vs full-scan map	Map fidelity, range-binned
Traversability precision / recall / F1	The field's standard measure
Positive obstacle detection recall	Core safety
Negative obstacle detection recall	Our headline — this is the contribution
Time-to-detect, in frames	The latency that actually matters
Per-frame latency (ms) on named hardware	SWaP — DRDO's own vocabulary
Points processed per frame	The budget axis
Minimum detectable obstacle in braking corridor	The provable safety number

The two charts that win

One — task metric against point budget. Ours dominating every baseline, converging on full-scan performance somewhere near 25–35% of the points.

Two — the failure case. Uniform decimation at 30% budget misses a ditch. Ours catches it. Same frame, same budget, side by side.

THE NUMBER WE ARE CHASING

“Matched obstacle-detection recall at 3× fewer points and 2.4× lower per-frame latency.” A ratio, a task metric, a named platform. Never *“significantly fewer.”*

04 — VALIDATION

Checked against what actually wins SIH

This plan was built backwards from past winners, not decorated with them afterwards.

WHAT THE WINNERS DID	OUR EQUIVALENT
Solar Masters — a measured 20% efficiency gain over fixed panels	Budget-versus-recall curve with a hard ratio
Team NYX , 2025 champion — section throughput for Railways	Points per frame, and maximum safe speed
Team Mad Astra — accuracy and scalability on NALCO's real process	Real off-road UGV data, not a toy scene
Every winner — a live working demo over slideware	Side-by-side replay demo, fully offline-capable
Winners read the sponsor's own domain	Negative obstacles, braking corridors, SWaP
Winners kept scope narrow and went deep	One algorithm, characterised properly

05 — SCOPE

What we deliberately do not build

This goes on a slide. Volunteering your boundaries reads as confidence and kills the gotcha before it is asked.

- **SLAM / localisation.** We *consume* ego-motion — RELLIS-3D ships GPS/IMU, simulation gives it free. Estimating it is a solved, separate problem.
- **Sensor fusion.** Lidar only. Adding cameras would confound the exact measurement we are making.
- **Path planning.** We define the braking corridor. We do not plan routes.
- **Dashboards, cloud, accounts.** Nobody asked for them.
- **Hardware.** This is a software problem statement.

Each is a deliberate decision with a reason attached. Always say the reason.

06 — SUBMISSION

The six slides

Maximum six slides, PDF. Use the official portal template if one is published — a non-conforming deck can be rejected on format alone.

SLIDE	CONTENTS
1	Title. Team Raysense · PS SIH26053 · full title · DRDO · Smart Automation · Software. Nothing else.
2	Problem. Open in DRDO's language — SWaP-constrained perception on UGVs in unstructured terrain. Then the gap in one sentence and the ambiguity diagram.
3	Solution. The single-pass loop as one clean diagram: need map → safety floor → allocation → acquisition → update → disambiguation.
4	Technical approach. Three sensor backends. The $\Delta\theta \leq h/r$ geometry with the speed table. Stack and datasets.
5	Feasibility & impact. The budget curve, the headline ratio, the max-safe-speed output, the path onto DRDO's existing AUGV programme.
6	References & team. Cite the prior art properly and prominently. Then one line: what is ours.

SLIDE 6 IS NOT A FORMALITY

It is where we win the novelty axis — by proving we read the field instead of pretending it does not exist.

07 – PRESENTATION

Winner mentality

Nine rules. Worth more than any feature we could add in the same time.

01 **Open in their language, not ours.**

The
first
thirty
seconds
should
sound
like
a
capability
briefing,
not
a
college
project.
*"UGVs
in
unstructured
terrain
operate
under
a
hard
SWaP
budget..."*
—
never
*"Hi,
we're
Raysense
and
we
made
a
cool
lidar
thing."*

02 Preempt the novelty question yourself.

Do
not
wait
to
be
caught.
*"Adaptive
lidar
sampling
is
established
—
AEye
ships
it,
NEC
published
it in
2020,
there's
a
2025
paper
on
temporal
cues.
What
does
not
exist
is..."*
This
flips
our
biggest
risk
into
a
credibility
asset.
Technical
juries
relax
visibly
when
a
team

knows
its
own
field.

03 **Lead with the failure, not the success.**

Show
naïve
sampling
missing
the
ditch
first.
Let
them
feel
it.
Then
show
ours.
Almost
no
team
structures
a
pitch
this
way,
and
it is
the
most
persuasive
order
available.

04 One number, repeated three times.

Slide

1,

the

demo,

the

closing

line.

Judges

see

twenty

teams

and

retain

exactly

one

number

per

team.

Choose

it

deliberately.

05 **Treat rounds 1–3 as free consulting.**

The
finale
has
four
stages;
the
first
three
are
mentoring
and
critique,
the
fourth
is
the
power
round.

Ask
each
juror:
*“What
would
you
need
to
see
to
trust
this
in
the
field?”*

Write
it
down
where
they
can
see
you
writing
it.
Then
ship
it
and

open
the
next
round
with
*“you
asked
for
X
—
here
it
is.”*
Most
teams
waste
these
rounds
defending
their
original
design.

06 **Never say “we invented.”**

Say
*characterised,
constrained,
quantified,
made
deployable.*
Overclaiming
is
the
fastest
way
to
lose
a
technical
jury.

07 **Bring offline fallbacks.**

Recorded
demo
video,
seeded
local
data,
zero
network
dependency,
spare
laptop,
spare
cable.
Teams
lose
rounds
to
laptops,
not
to
logic.

08 **One presenter — but everyone answers for their own module.**

Juries
probe
by
asking
the
quiet
member
a
question.
Every
person
must
be
able
to
defend
their
component
unprompted.

09 Volunteer the limitations before you're asked.

*"Here's
what
we
don't
handle,
and
here's
why
that's
the
right
scope."
Confidence,
not
weakness.*

The hard questions, and our answers

THEY WILL ASK	WE ANSWER
Which sensor does this run on?	Three backends. Steerable if you have it, masking if you don't, replay for evaluation. Agnostic by design.
How is this different from AEye, or the 2025 paper?	Those are urban and unconstrained. Ours is off-road, negative-obstacle-aware, and carries a provable safety floor.
Isn't this just simulation?	No. Primary results are on RELLIS-3D — real lidar from a real off-road UGV. Simulation is only for controlled negative-obstacle ground truth.
Does it actually save battery?	On a spinning sensor, no laser energy — bandwidth, memory and compute. On a steerable sensor, laser energy too. Here is the measured number.
What if it misses something?	Below the floor it cannot. $\Delta\theta \leq h/r$ is enforced inside the braking corridor before any adaptive allocation. Here is the minimum detectable size.
Is it real-time?	X ms per frame on [named platform]. Here is the profile.
Why not just scan the front densely?	That is baseline 4. Here is the chart where we beat it — threats aren't only ahead, and staleness matters.

08 — EXECUTION

The schedule

Scheduled against a **20 September** idea submission. Phases overlap deliberately — the curve must exist early, even ugly.

PHASE 0 Unblock

Days 0–1 · 5–6 Sep

- Confirm the real deadline — 20 or 30 September — on sih.gov.in
- Get the internal hackathon date from the SPOC. This is the deadline that binds.
- Pull the full official PS text and map **every clause** to a deliverable
- Confirm six members, same college, at least one female member — eligibility gate
- Start the RELLIS-3D download. It is large.
- Assign roles

PHASE 1 Foundation

Days 1–5

- RELLIS-3D loader; render one scan
- Full-scan 2.5D elevation map plus viewer — this becomes ground truth for everything
- Read the three papers: NEC MEMS adaptive lidar · Adaptive LiDAR Scanning 2025 · RELLIS-3D
- Budget harness: subsample a recorded scan to B points, rebuild the map, score it

PHASE 2 Baselines and first curve

Days 4–9

- Baselines 1–4 implemented
- Metrics: elevation RMSE, traversability P/R/F1, obstacle recall
- **First budget-versus-quality curve.** Even ugly — this is the project's spine.

PHASE 3 The allocator

Days 7–12

- Need map, all six terms
- Safety floor with the $\Delta\theta \leq h/r$ geometry and max-safe-speed output
- Single-pass predictive loop with ego-motion warping
- Ours on the curve, beating the baselines

PHASE 4 The contribution

Days 10–14

- Ray-level UNKNOWN versus EMPTY accounting
- Negative-obstacle signature detection
- CANDIDATE_NEGATIVE priority budgeting
- **Find and capture the failure case** — naive misses a ditch, ours catches it
- Simulated scene with planted negative obstacles and exact ground truth

PHASE 5 Submit

Days 12–15

- Side-by-side demo, offline-capable
- Both headline charts finalised
- The six-slide PDF
- Internal hackathon presentation — and rehearse the nine rules

PHASE 6 Harden, if nominated

Oct–Nov

- Jetson-class benchmark for real SWaP numbers
- Closed-loop simulation · SemanticKITTI generalisation
- Learned need map — stretch goal only
- Rehearse the power round as a deliverable separate from the code

PHASE 7 Grand finale

December

- 36 hours. **Integrate nothing new.**
- Ship the jury's feedback between rounds. That is the whole game.

09 — TEAM

Roles

Six members, non-overlapping ownership. Everyone defends their own module to the jury.

#	ROLE	OWNS
1	Lead / presenter	Narrative, PS clause mapping, jury interaction, the nine rules
2	Mapping	2.5D quadtree, fusion, traversability output
3	Allocator	Need map, budget optimisation, safety floor
4	Evaluation	RELLIS-3D pipeline, baselines, metrics, the two charts
5	Simulation	Closed loop, planted negative obstacles, ground truth
6	Demo	The side-by-side viewer — what judges actually look at

The demo role is not junior work. It is the highest-visibility component in the room.

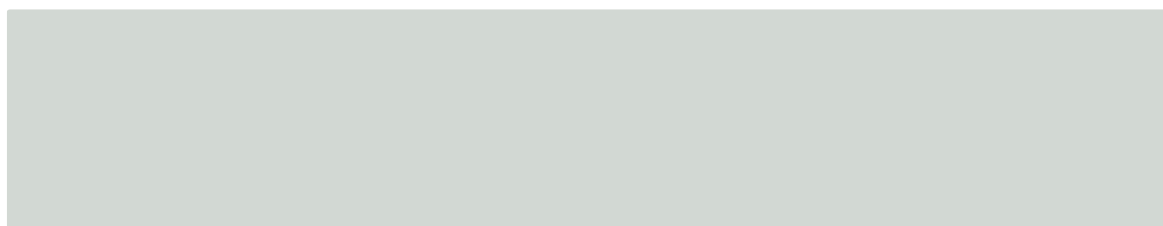
10 – RISK

What could kill this

RISK	SEVERITY	MITIGATION
Deadline is the 20th, not the 30th	CRITICAL	Verify day 0. This plan already assumes the 20th.
No registered SPOC at our college	CRITICAL	Verify day 0. Nothing else matters if this fails.
RELLIS-3D too large or slow to process	HIGH	Start day 0. Work one sequence first.
Negative obstacles are rare in real data	HIGH	Exactly why we plant them in simulation. The two-leg design handles this.
Allocator doesn't beat the front-focused baseline	HIGH	Report it honestly and analyse why. An honest negative result beats a tuned lie — juries can smell the difference.
Scope creep	MEDIUM	Section 05 is a contract. Re-read it weekly.
Demo fails live	MEDIUM	Offline data, recorded video, spare laptop.
Judged as “not novel”	MEDIUM	Rule 2. Preempt it in our own opening.

CLOSING

The three things that matter most



01

**Verify the deadline
and the SPOC today**

Over 80% of SIH teams are eliminated on administration, before anyone writes a line of code. Do not be one of them.

02

**Get the curve working
early, even ugly**

Budget versus recall is the spine of the entire project. Everything else is decoration hung on it.

03

**Find the failure case,
and show it first**

Naive adaptive sampling driving into a ditch is the most persuasive ten seconds we will ever have in front of a jury.